

Orthemology notation gallery

Status: DRAFT — not peer reviewed. Rendered by the R7B mathematical- typesetting pipeline (Decision 0023). Every normative symbol in `docs/notation-registry.yaml` appears below as real mathematics: GitHub renders the `...$` inline math and fenced `math` blocks through MathJax, and `scripts/build_pdfs.py` renders `artifacts/notation-gallery.pdf` through the pinned Typst compiler via `scripts/latex.to.typst_math.py`. This is a **typography layer, not a notation redesign**: each row’s meaning is the registry role, unchanged (Decision 0005). `scripts/validate_math_source.py` asserts a bijection between non-retired symbols and this gallery and that `scripts/validate_pdf_math.py` finds no missing-glyph (notdef/tofu) in the rendered PDF — the defect reproduced in `docs/project-closure/r7b/R7B-PDF-MATH-BASELINE.md` (the core μ -vector and $\mathbf{C}$ -vector rendering as notdef under the old `#raw` path). Below they compose correctly as $\vec{\mu}$ and $\vec{C}$.

1. Normative symbols rendered as mathematics

Registry key: A
Rendered: A
Role: declared analysis (identifier + version)
Registry key: alpha, beta, ...
Rendered: α, β
Role: actor indices
Registry key: T = task(A)
Rendered: $T = \text{task}(A)$
Role: task component of the analysis
Registry key: M_univ, O_univ
Rendered: $\mathcal{M}, \mathcal{O}$
Role: universal occurrence and ortheme universes
Registry key: M_A, O_A
Rendered: $\mathcal{M}_A \subseteq \mathcal{M}, \mathcal{O}_A \subseteq \mathcal{O}$
Role: analysis-active domain and repertoire
Registry key: Inst_A
Rendered: $\text{Inst}_A \subseteq \mathcal{M}_A \times \mathcal{O}_A$
Role: analysis-relative instantiation (the primitive)
Registry key: O*(m; A)
Rendered: $O^*(m; A)$
Role: true (actual) profile of m under A
Registry key: Pi_A
Rendered: Π_A
Role: complete-profile space under A
Registry key: Pi_A_partial
Rendered: Π_A^∂
Role: partial-profile space under A
Registry key: C_hat
Rendered: $\hat{C}_{A, \alpha, t}(m)$
Role: candidate set of complete profiles
Registry key: p_hat
Rendered: $\hat{p}_{A, \alpha, t}(m) \in \Pi_A^\partial$

Role: inferred (partial) profile — belief, never ground truth
Registry key: Omega
Rendered: $\Omega_{A, \alpha, t}$
Role: observation map
Registry key: mu
Rendered: $\mu = \langle g, S_\mu, \text{select}_\mu, \text{prov}, \text{ver} \rangle$
Role: metaortheme type
Registry key: pi_mu
Rendered: π_μ
Role: paired meta-policy
Registry key: mu_bar
Rendered: $\bar{\mu}_{e, j}$
Role: the j-th metaorthemma in episode e
Registry key: MetaInst
Rendered: $\text{MetaInst}(\bar{\mu}_{e, j}, \mu)$
Role: token typing of a metaorthemma under a metaortheme
Registry key: Gamma_E
Rendered: $\Gamma_E = (E, \rightsquigarrow)$
Role: episode DAG
Registry key: e_Gamma
Rendered: $e_\Gamma = \text{comp}(\Gamma_E)$
Role: composed boundary-level episode
Registry key: GoalSchema
Rendered: $\text{GoalSchema}(\alpha)$
Role: parametric goal schema
Registry key: G_target
Rendered: $\mathcal{G}_{\alpha, A_\alpha}$
Role: actor-alpha’s grounded target profile set
Registry key: Rep_A
Rendered: Rep_A
Role: representation family under A
Registry key: chi
Rendered: $\chi \in \text{Rep}_A$
Role: one representation

Registry key: L.A_star
Rendered: $L_A^*(\chi)$
Role: best attainable loss/risk under representation χ
Registry key: select_mu
Rendered: select_μ
Role: metaortheme state-selecting evidence procedure
Registry key: estatus_e
Rendered: estatus_e
Role: per-claim evidence-status map of episode e
Registry key: eps_A
Rendered: ϵ_A
Role: accepted tolerance of the analysis
Registry key: theta_stop
Rendered: θ_{stop}
Role: ANDON stop/escalation risk threshold
Registry key: K_A
Rendered: $\mathcal{K}_A$
Role: cause repertoire
Registry key: R_A
Rendered: $\mathcal{R}_A$
Role: route repertoire
Registry key: W_A
Rendered: $\mathcal{W}_A$
Role: warrant-state repertoire
Registry key: Q_e
Rendered: $\mathcal{Q}_e$
Role: claim ledger of episode e
Registry key: delta_e
Rendered: δ_e
Role: residual-disposition map of episode e
Registry key: ReqPath
Rendered: $\text{ReqPath}(e)$
Role: governance-derived required pathway verdicts

2. Structural constructs (typesetting showcase)

The pipeline handles the full range of publication constructs the corpus uses. These are the constructs named in the Decision 0023 design spike.

2.1 Episode signature (display)

The formal-core episode signature — the exact site of the reproduced notdef defect — rendered as real mathematics (the vectors $\vec{\mu}$, $\vec{C}$ compose):

$$e = \langle \text{id}; m, \kappa, v; x, H; \alpha, w, A, T, \\ t; \vec{\mu}, \text{MetaTok}, \pi; \vec{C}, \hat{p}; r; \\ \text{estatus}; \mathcal{Q}; \delta; \\ \text{hand_in}, \text{hand_out}; a, \\ \text{Succ} \rangle$$

2.2 Multi-line aligned equations

$$\begin{aligned} \text{TokenAdequate}(\bar{\mu}, e) &\Leftrightarrow \text{MetaInst}(\bar{\mu}, \mu) \\ &\quad \wedge \text{Compatible}(\bar{\mu}, A(e)) \\ \text{V3c}(e) &\Leftrightarrow \\ &\quad \forall \bar{\mu} \in \text{MetaTok}(e) : \\ &\quad \quad \text{TokenAdequate}(\bar{\mu}, e) \end{aligned}$$

2.3 Set comprehension and predicate

Grounded target (a set of **profiles**, $\subseteq \Pi_{A_\alpha}$, not of occurrences): $\mathcal{G}_{\alpha, A_\alpha} = \{p \in \Pi_{A_\alpha} \mid \text{GoalSchema}(\alpha)(p)\}$, and strict soundness $\text{StrictlySoundReasoning}_\chi(e) := \text{ReasoningPathAdequate}_\chi(e) \wedge \text{TokenTruthLinked}_\chi(e)$.

2.4 Underbrace

$$\underbrace{O^*(m; A)}_{\text{ground truth, never asserted from discourse}} \neq \underbrace{\hat{p}_{A, \alpha, t}(m)}_{\text{inferred belief}}$$

2.5 Labeled arrows (world transition vs learner update)

$m_t \xrightarrow{a_t} m_{t+1}$ (occurrence lineage) is not $\theta_t \xrightarrow{U_A(e_t)} \theta_{t+1}$ (learner update).

2.6 Table containing inline math

Object	Symbol	Non-claim
observation map	$\Omega_{A, \alpha, t}$	not the occurrence
inferred profile	$\hat{p}_{A, \alpha, t}(m)$	not $O^*(m; A)$
metaorthemma	$\bar{\mu}_{e, j}$	not the metaortheme type μ

All notation meanings are those of `docs/notation-registry.yaml`; nothing here introduces a new symbol or claim.